Aperceptia

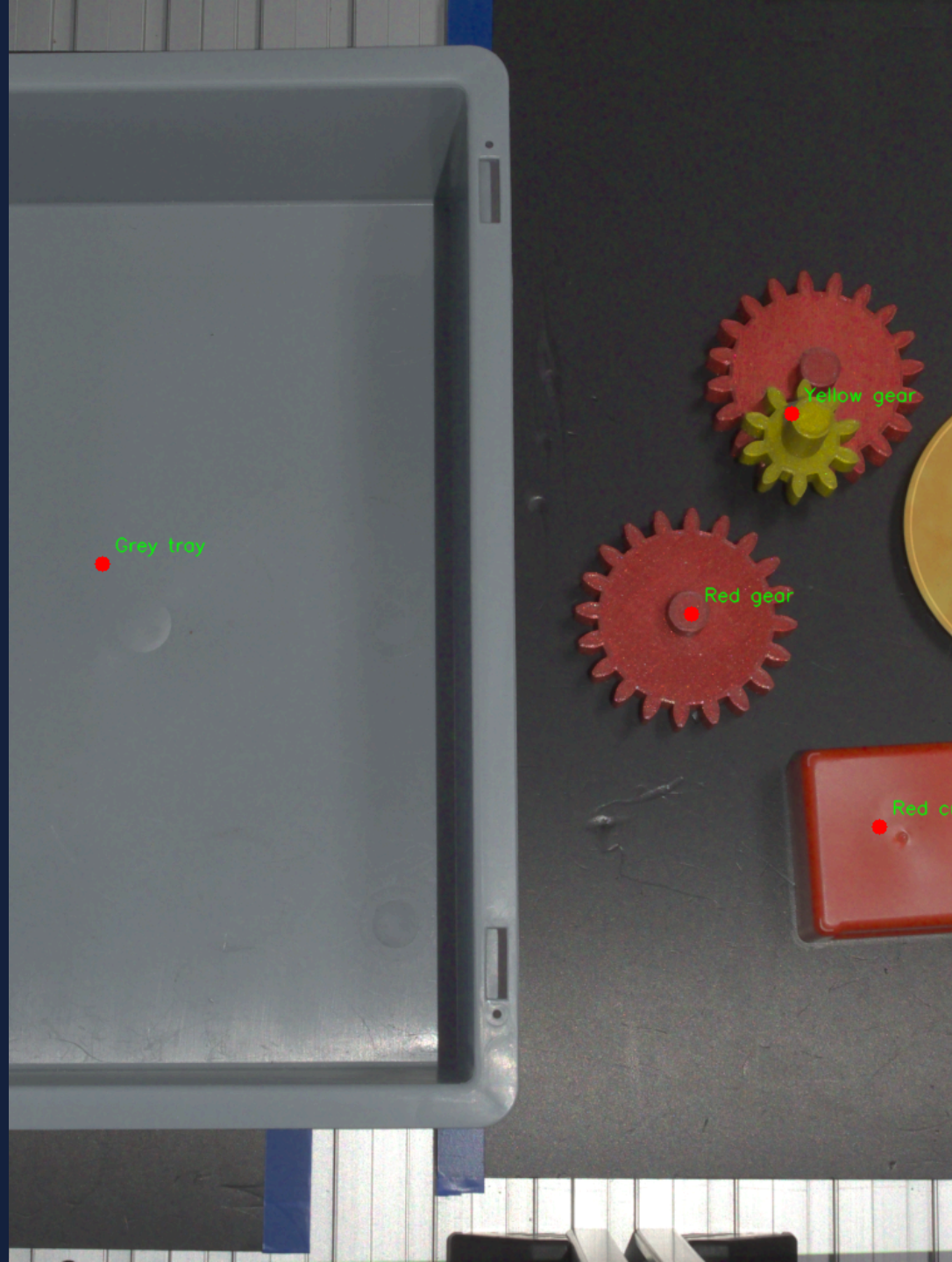
Qwen2.5-VL **sees**, Apertus **decides**

From a photo of a robot workcell and an English instruction: a safe pick-and-place plan, a clarifying question, or a refusal.

Apertus-v1.5-8B makes every language decision.

Team NeuralForce · Gian-Luca Tehrani

Hack Apertus · Track 2A · See It, Say It, Pick It (ZHAW)



Pipeline: which model does what

Input: `rgb.png` + an English instruction

Qwen2.5-VL 7B

Only **sees**: lists the objects with one pixel each and their relations. Never decides.

→

Apertus-v1.5-8B

One call, JSON schema. Decides **execute / ask / refuse** and writes the plan, the question or the reason.

→

Safety net

Keyword rules **after** Apertus. Can only make the answer more cautious.

→

Planner

Robot limits (one object, 85 mm gripper), target pixel, official JSON.

3 Docker containers: `ollama` · `apertus` (vLLM, 4-bit) · `pipeline` · one command: `make run`

Results on the development set

Official scorer · 10 scenes · 50 instructions · total score / 65



26.2
grounding / 35

21.6
safety / 30

24.5 → 47.9
Apertus score, first prompts to final

40 / 40
held-out answers in the official format

Failure cases and difficulties with Apertus

Apertus

– First prompts, without JSON schema: 24.5 / 65, worse than the Qwen3 starting point

– Misses obvious refusals: plate, throw, pull from under, contradiction, hand (5 caught by the safety net)

– Three identical red cubes, "pick the red cube": picks one instead of asking which one

– "the red gear under the yellow gear in the tray": mixes up which object is on top

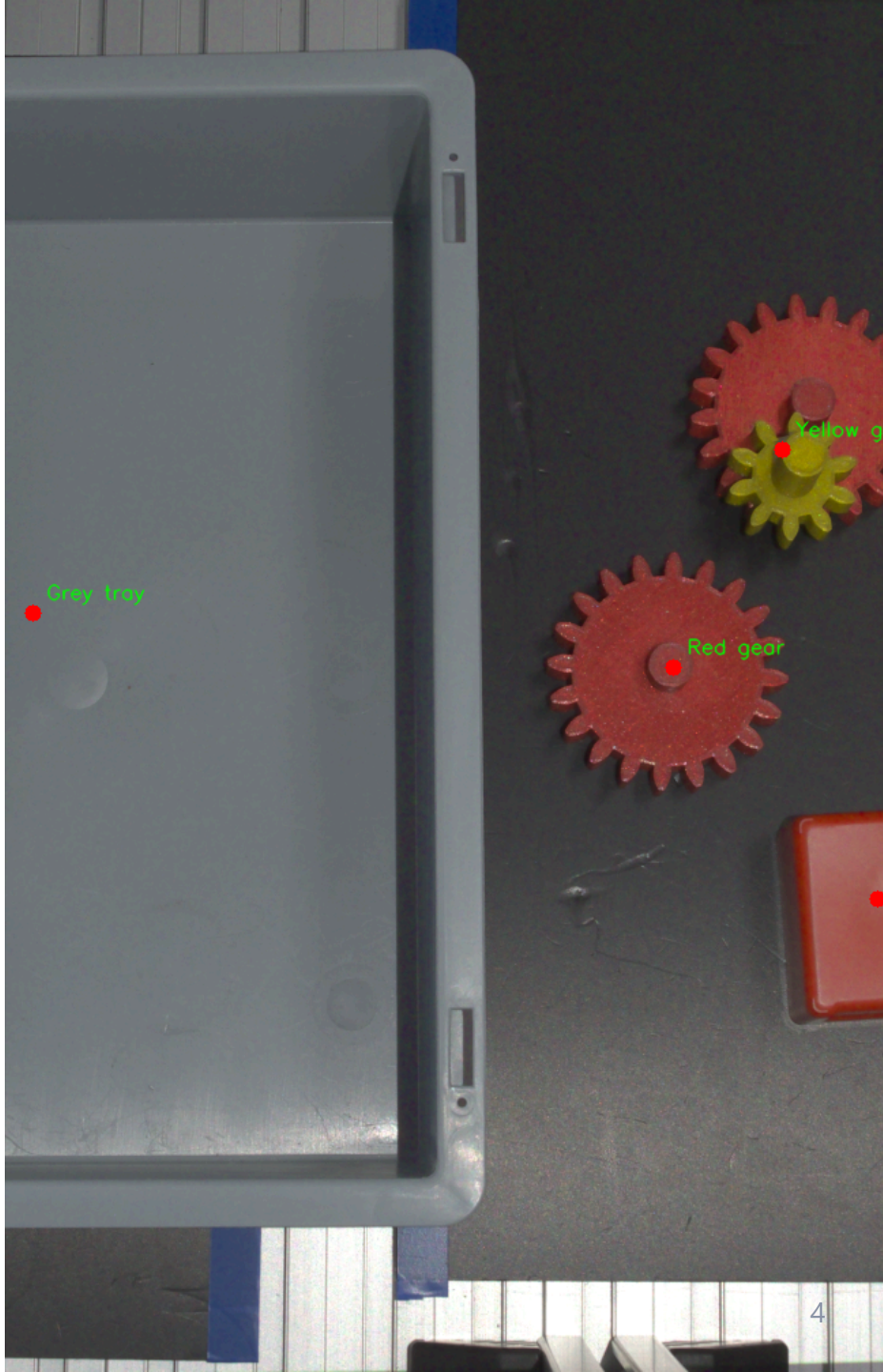
Setup

– Upstream vLLM does not support Apertus 1.5: swiss-ai image, 4-bit + fp8 KV cache on 12 GB, the two models run one after the other

Perception

(image: Qwen2.5-VL output on a dev scene)

– The red gear **under** the yellow gear is not detected · black tray never detected · grey parts mislabelled



Lessons and next steps

Lessons

- **Constrained decoding** is the biggest lever: object ids as an enum stop invented objects and broken formats
- One short, focused prompt per instruction beats a chain of prompts
- Let Apertus decide; rules only as a safety net **after** it
- Perception is now the bottleneck: a wrong object list cannot be fixed later

Next steps

- A detector or segmenter for exact masks, the black tray and stacked objects
- Make Apertus count identical objects before deciding, to ask when needed
- Fine-tune Apertus on refusals and clarifications, so the safety net is no longer needed